

Of Errors and Echoes: Responsibility-Aware Credit Assignment for Tool-Use Agents

Xingjian Diao^{1*}, Tianyu Yang^{2*}, Wenjun Huang³, Xiangchi Yuan⁴, Chunhui Zhang¹
Xingchen Zhao⁵, Weiyi Wu¹, Haotian Xu⁶, Soroush Vosoughi¹,
Xiangliang Zhang^{2†}, Jiang Gui^{1†}

¹Dartmouth College ²University of Notre Dame ³University of California, Irvine
⁴Georgia Institute of Technology ⁵Northeastern University ⁶Independent Researcher
xingjian.diao.gr@dartmouth.edu, tyang4@nd.edu

Abstract

Reinforcement learning improves tool-use agents, yet fine-grained credit assignment remains challenging even with step-level feedback. An observed error may be inherited from an earlier decision, while multiple responsible decisions may share the same downstream consequence or participate in later recovery. We introduce Error-Chain-aware Optimization (ECHO), a responsibility-aware credit assignment framework that separates responsibility identification from credit allocation. A frozen judge combines retrospective trajectory analysis with decision-time assessment from the policy-visible prefix to distinguish newly introduced errors, inherited consequences, missed corrections, and successful corrections. ECHO then counts shared downstream consequences once across responsible decisions and constrains rewards for repairing self-induced errors through source-funded recovery, producing bounded action-level supervision alongside terminal supervision. Experiments on six diverse tool-use benchmarks at both 4B and 8B scales show strong and consistent improvements over the compared baselines, with ECHO achieving the highest score on five of six benchmarks at each scale. These results support decision responsibility as a useful basis for fine-grained credit assignment in tool-use agents. Our code is available at <https://anonymous.4open.science/r/OctARR-ECHO>.

1 Introduction

Tool-use agents retrieve information and modify environments through sequential decisions (Yao et al., 2022; Schick et al., 2023; Qin et al., 2024). Reliable execution requires tracking state changes and constraints that shape subsequent tool calls (Zhou

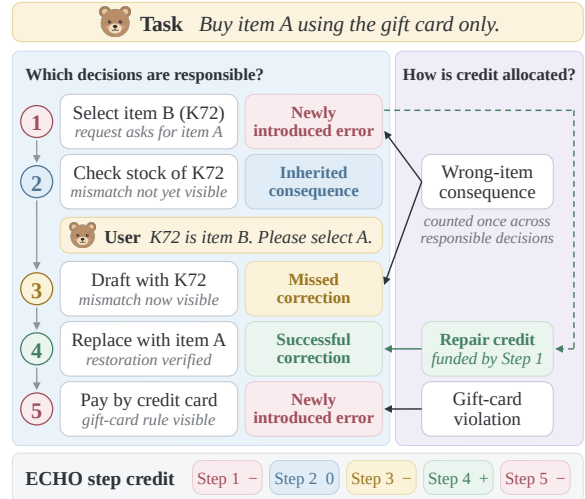


Figure 1: Illustrative error chain showing ECHO’s distinction among newly introduced errors, inherited consequences, missed corrections, and successful corrections, with shared consequences allocated without double counting.

et al., 2024a; Yao et al., 2024; Lu et al., 2025). Reinforcement learning can improve task performance, but terminal rewards reveal only whether a trajectory succeeds or fails, leaving unresolved which decisions are responsible for the outcome (Wang et al., 2025b; Qian et al., 2026). Prior work therefore introduces step-level or process-level feedback (Lightman et al., 2024; Wang et al., 2024; Zhang et al., 2025b; Zheng et al., 2025). However, finer-grained supervision does not by itself resolve credit assignment. An observed error may be inherited from an earlier decision, while multiple decisions may share the same downstream consequence or participate in later recovery. These dependencies raise two questions: which decisions are responsible, and how should credit be allocated among them?

Figure 1 illustrates these two questions. The user requests item A and specifies that payment must use the gift card only. The agent first selects the wrong item B and passes its identifier K72 down-

*Equal contribution, listed in random order.

†Corresponding authors.

stream. Step 2 queries inventory using this identifier, even though it refers to the wrong item B. The user later reveals that K72 refers to B rather than A, yet Step 3 still fails to correct the mistake. Step 4 restores the intended item A, while Step 5 independently violates the payment constraint by using a credit card instead of the gift card. The trajectory therefore contains error introduction, propagation, missed correction, and recovery, raising two distinct credit-assignment questions.

First, which decisions should be held responsible? For propagated errors, the key distinction is whether the discrepancy was identifiable when the action was taken. Step 2 (*inventory query with K72*) inherits an erroneous state, but the mismatch is not visible in its local context. Step 3 (*continued use after disclosure*), by contrast, ignores a mismatch that has already become visible and thus constitutes a missed correction. Responsibility must therefore be assessed from decision-time information, rather than from the erroneous state alone or from evidence revealed later.

Second, once responsible decisions are identified, how should credit be allocated? Correctly identifying responsible decisions is still not enough: their credit cannot always be assigned independently. Step 1 (*wrong item selection*) and Step 3 (*missed correction*) both contribute to the same wrong-item consequence, so assigning each the full penalty would double-count the same impairment. Step 4 (*item repair*) introduces a different challenge: recovery may deserve positive credit, but not in a way that rewards the policy for first creating its own repair opportunity. Step 5 (*payment violation*), meanwhile, is an independent defect. Credit assignment must therefore account for shared consequences and recovery, rather than scoring each responsible step independently.

We introduce **ECHO** (**E**rror-**C**hain-aware **O**ptimization), which addresses these two problems jointly by assigning credit over error chains rather than treating steps independently. ECHO distinguishes four responsibility events: ① newly introduced errors (NEW), ② inherited consequences (INH), ③ missed corrections (MISS), and ④ successful corrections (CORR). ECHO first performs responsibility identification. A frozen trajectory judge uses the complete trajectory to propose candidate events and establish propagation and restoration evidence, while the local responsibility of each selected action is independently assessed using only its policy-visible prefix. In Figure 1, Step

2 (*inventory query with K72*) is treated as an inherited consequence, whereas Step 3 (*continued use after disclosure*) is identified as a missed correction. Later observations can therefore verify propagation and restoration without leaking future information into the responsibility judgment of an earlier decision.

ECHO then performs credit allocation. When multiple responsible decisions contribute to the same downstream consequence, shared-consequence accounting counts that consequence once and allocates it across supported sources of responsibility. For repairs of self-induced errors, source-funded recovery constrains positive correction credit by the budget supplied by the corresponding source error. Accordingly, Step 1 (*wrong item selection*) and Step 3 (*missed correction*) are not both charged the full wrong-item consequence, while the repair credit at Step 4 (*item repair*) is bounded by its originating error. These responsibility signals are converted into bounded action-level advantage corrections and optimized alongside the original terminal supervision.

We evaluate ECHO at 4B and 8B scales across six tool-use benchmarks. At both scales, ECHO ranks first among the compared models on five of six benchmarks. Further ablation and frozen-judge analyses examine the roles of responsibility identification, credit allocation, and judge quality.

Our contributions are threefold:

- We formulate local supervision through decision responsibility, separating error sources, inherited consequences, and correction behavior.
- We develop ECHO to connect independent prefix assessment with bounded consequence and recovery accounting for fine-grained policy optimization.
- We establish cross-scale and cross-benchmark evidence and examine the effects of error selection and frozen-judge choice through targeted comparisons.

2 Related Work

Tool-Use Agent Learning. Reinforcement learning has substantially improved tool-use agents through reward design, multi-turn rollouts, and simulated users (Qian et al., 2026; Feng et al., 2026a; Zhao et al., 2025; Dong et al., 2026; Zhou et al., 2024b; Wen et al., 2024; Luo et al., 2025), as well as through scaled training environments (Song et al., 2026; Wang et al., 2026b, 2025a; Li

et al., 2025). These methods primarily optimize trajectory-level outcomes or scale the environments used for training. ECHO is complementary, addressing how supervision is distributed across individual agent decisions when errors propagate, become observable, or are later corrected.

Process Supervision and LLM Critics. Step-level feedback provides denser supervision than terminal rewards (Lightman et al., 2024; Wang et al., 2024; Zhang et al., 2025b; Zheng et al., 2025), and reward models and LLM critics extend this idea to search and tool-use agents (Xia et al., 2025; Li et al., 2026b; Xi et al., 2026; Tan et al., 2025, 2026; Zhou et al., 2025; Zhang et al., 2026). Such signals localize problematic steps, but localization alone does not establish decision responsibility: an observed error may be inherited from an earlier decision, and judging a decision with later evidence can penalize it for information unavailable when it was made. ECHO therefore uses the full trajectory only to propose candidate events and verify propagation or restoration, assessing each decision independently from its policy-visible prefix.

Credit Assignment and Error Attribution. Beyond identifying responsible decisions, a complementary question is how delayed outcomes should be allocated among them. Reward redistribution and recent agent methods estimate step or turn contributions from delayed returns, trajectory structure, counterfactual rollouts, or turn-level value estimates (Arjona-Medina et al., 2019; Meulemans et al., 2023; Zhang et al., 2023; Feng et al., 2026b; Li et al., 2026a; Wang et al., 2026a; Gandhi et al., 2026; Meng and Chen, 2026; Tao et al., 2026; Liao et al., 2026; Zhan et al., 2026), while failure-attribution studies identify root-cause errors for post-hoc diagnosis (Zhang et al., 2025a; Zhu et al., 2025). Conceptually related to structural accounts of responsibility (Chockler and Halpern, 2004), ECHO explicitly separates responsibility identification from credit allocation: shared-consequence accounting counts each consequence once across its responsible decisions, and source-funded recovery bounds self-induced repair credit by the budget of its originating error.

3 ECHO Pipeline

ECHO complements the terminal reward with action-level corrections δ_t (Figure 2, Algorithm 1). A frozen judge classifies selected actions as NEW, INH, MISS, or CORR using decision-time context

Algorithm 1 One ECHO update

Require: policy π_θ , frozen judge $\mathcal{J}$, batch of tasks

- 1: Sample G trajectories per task; record R_i and $h_{i,t}$
- 2: **for** each trajectory i **do**
- 3: Propose events, links, and restorations from the full trajectory
- 4: **for** each selected action $t (\leq K)$ **do**
- 5: Assess type z from decision-time context only
- 6: **end for**
- 7: Retain events passing structural checks and g_e
- 8: Allocate and weight events Eqs. (3)–(6)
- 9: Normalize into budgets b_e^s Eq. (7)
- 10: Cap self-induced repair credit Eqs. (8)–(9)
- 11: Aggregate corrections $\delta_{i,t}$ Eq. (10)
- 12: **end for**
- 13: Compute A_i^{out} and optimize J_{ECHO} Eqs. (11)–(15)

(§3.2); ECHO then counts each shared penalty once, caps self-induced repair credit by its source penalty (§3.3–3.4), and optimizes δ_t alongside the terminal advantage (§3.5). In Figure 1, Steps 1 and 3 split the wrong-item penalty, Step 5 is charged separately, Step 2 receives none, and Step 4’s repair credit is funded by Step 1’s penalty.

3.1 Traces and Responsibility Events

Trajectory i of policy π_θ on task x consists of actions $a_{i,t}$, observations $o_{i,t}$, and terminal reward R_i . The decision-time context $h_{i,t}$ records the information visible to the policy before $a_{i,t}$, after any context truncation or memory retrieval. We represent checkable task requirements as predicates $f \in \mathcal{F}_x$, such as “payment = gift card” in Figure 1. The judge outputs responsibility events

$$e = \langle t, f, d, z \rangle, \quad (1)$$

where t is the assessed action, f the affected predicate, d the deviation origin (self-induced, authenticated external, or unknown), and z the responsibility type. Each event also stores supporting evidence pointers and, for a missed correction, a feasible alternative response (§A). A single action may produce multiple events, for example by repairing one predicate while violating another; their action-level coefficients are aggregated in Eq. (10).

3.2 Evidence-Grounded Responsibility Assessment

Responsibility types. Local responsibility is assessed from evidence visible in $h_{i,t}$. A newly introduced error (NEW) is a distinct decision defect evident from a fact or constraint visible to the policy. A missed correction (MISS) requires an existing deviation, visible evidence of the discrepancy, and

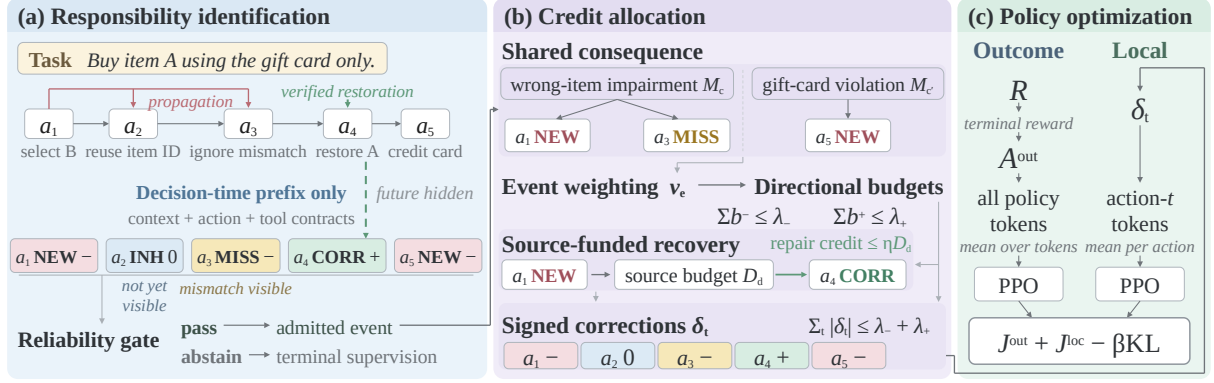


Figure 2: Overview of ECHO. (a) Full-trajectory review proposes candidate events, propagation links, and restoration evidence, while decision-time assessment classifies each selected action as NEW, INH, MISS, or CORR. (b) Shared consequences are counted once across responsible actions, then weighted, normalized, and adjusted by source-funded recovery to yield signed action-level corrections δ_t . (c) Terminal and local supervision are optimized through separately clipped outcome-level and action-level branches with KL regularization.

a feasible permitted response that the action fails to take. An inherited consequence (INH) reflects an earlier deviation with explicit provenance but involves neither a new defect nor a missed correction. A successful correction (CORR) is a legitimate response to a visible deviation and additionally requires execution evidence confirming restoration of the predicate or a required precondition; unverified repair claims receive no positive credit. Actions without events retain only terminal supervision, and for external failures ECHO assesses only the agent’s response to the incident. Duplicate events are merged, while ambiguous or conflicting labels resolve to a single supported type or abstention.

Independent prefix assessment. The judge plays two roles. Given the full trajectory, it proposes candidate events, propagation links, and restoration evidence. For at most K selected actions, it then independently assesses responsibility from $(h_{i,t}, a_{i,t})$ and the tool contracts visible at that time, without access to the proposals, hidden state, or future observations. Under fixed decoding, the local assessment of a selected action is therefore invariant to the trajectory’s suffix. Hindsight informs candidate selection, propagation links, restoration evidence, and event weighting, but not the decision-time responsibility judgment. Of the $K = 8$ slots, six are assigned to proposed actions and two to non-proposed sentinel actions (§A).

Selective supervision. Each event must pass structural checks on its evidence pointers, visibility, temporal order, provenance, and nonempty policy span. A binary gate g_e further admits only evidence strata whose semantic error rate, estimated from

independent annotations, falls below a calibrated bound (§B). Events that fail either check receive only terminal supervision.

3.3 Shared Consequences and Propagation Weighting

Constraint-impairment proxy. Before inspecting outcomes, a fixed task adapter extracts nonredundant requirement predicates $\mathcal{F}_x$ from public instructions and tool contracts, with masses ω_f that sum to one and are uniform by default. Predicate versions and primitive subfacts share the mass of their underlying requirement; obligations outside this inventory contribute only through the direct-defect term below. Repeated manifestations of the same predicate’s impairment form one *consequence* c with mass

$$M_c = \omega_{f(c)} s_c, \quad \sum_c M_c \leq 1, \quad (2)$$

where severity s_c is 0 for unconfirmed, 1/2 for reversible, and 1 for terminal or irreversible impairment. Recovery does not erase the source defect or reduce its recorded maximum severity.

Divisible ownership. Let P_c contain the responsibility slots linked to consequence c , including uncertified slots. After reserving a fraction ζ_c for uncertain provenance, each slot receives an equal share:

$$a_{ec} = \begin{cases} \frac{1 - \zeta_c}{|P_c|}, & e \in P_c, |P_c| > 0, \\ 0, & \text{otherwise,} \end{cases} \quad (3)$$

where ζ_c is 0 for complete recorded provenance, 1/2 for unresolved additional sources, and 1 for unsupported or conflicting ownership. The reserved

fraction and the shares of rejected slots are not reassigned. Equal division is an accounting convention: in Figure 1, the source error at Step 1 and the missed correction at Step 3 each receive half, not all, of the wrong-item consequence.

Event weights. For a negative event,

$$\hat{\ell}_e^- = \kappa \iota_e + (1 - \kappa) \sum_c M_c a_{ec}, \quad (4)$$

where $\iota_e \in \{0, 1\}$ indicates a certified distinct decision defect and $0 < \kappa \leq 1$. The direct term counts each certified defect once, including defects that are later repaired. For a successful correction r , let first-restoration shares z_{rc} satisfy $\sum_r z_{rc} \leq 1$, and define

$$\hat{\ell}_r^+ = \kappa + (1 - \kappa) \sum_c M_c z_{rc}. \quad (5)$$

Let $\mathcal{F}_e \subseteq \mathcal{F}_x$ contain the predicates reached from event e through validated propagation links, namely explicit reuse of corrupted values or invalidated conditions. For corrections, $\mathcal{F}_e$ is restricted to the repaired origin's footprint. For either event direction,

$$p_e = \sum_{f \in \mathcal{F}_e} \omega_f, \quad v_e = g_e \hat{\ell}_e [1 + \alpha p_e], \quad (6)$$

where $\hat{\ell}_e$ denotes the corresponding negative-event or correction base weight and $0 \leq \alpha \leq 1$. Since $p_e \leq 1$, propagation increases an event's weight by at most a factor of $1 + \alpha$.

3.4 Source-Funded Recovery Budgets

Let $\mathcal{E}_i^-$ contain the admitted NEW and MISS events of trajectory i , and $\mathcal{E}_i^+$ its execution-verified CORR events. INH and abstained events receive no auxiliary budget and retain only terminal supervision. Event weights are normalized into directional budgets

$$b_e^s = \lambda_s \frac{v_e}{\max\{1, \sum_{u \in \mathcal{E}_i^s} v_u\}}, \quad s \in \{-, +\}, \quad (7)$$

so that each direction totals at most λ_s , while weight sums below one retain their original scale.

Source-funded recovery. Without further constraints, a policy could earn positive credit by repairing errors it introduced itself. ECHO therefore funds such repairs from the penalty assigned to

their source. For a self-induced origin d , the source penalty is

$$D_d = \sum_{e: z_e = \text{NEW}, d(e)=d} b_e^-, \quad (8)$$

with each NEW charge assigned to exactly one origin. Let $\mathcal{R}_d$ contain the eligible restorations of origin d , with total provisional credit $B_d^+ = \sum_{r \in \mathcal{R}_d} b_r^+$. For $B_d^+ > 0$, set $\gamma_d = \min\{1, \eta D_d / B_d^+\}$, and otherwise set $\gamma_d = 0$. The paid correction coefficients are

$$\tilde{b}_r^+ = \begin{cases} \gamma_d b_r^+, & d \text{ self-induced,} \\ b_r^+, & d \text{ authenticated external,} \\ 0, & \text{otherwise,} \end{cases} \quad (9)$$

where $0 < \eta \leq 1$ bounds self-induced repair credit to a fixed fraction of its source penalty. Partial repairs of the same origin share this budget, unused credit is discarded, and each authenticated external incident permits one recovery entitlement. In Figure 1, Step 1's penalty funds the repair at Step 4, while the payment violation at Step 5 is charged separately.

Action-level corrections. Aggregating events on each action yields

$$\delta_{i,t} = \sum_{r \in \mathcal{E}_{i,t}^+} \tilde{b}_r^+ - \sum_{e \in \mathcal{E}_{i,t}^-} b_e^-, \quad (10)$$

so that $\sum_t |\delta_{i,t}| \leq \lambda_- + \lambda_+$. The resulting $\delta_{i,t}$ enters the local objective (§3.5) without further rescaling. Positive and negative event records are retained even when their coefficients cancel on the same action.

3.5 Length-Balanced Policy Optimization

Let $q_{i,k}$ indicate a loss-eligible policy token and $m_{i,t,k}$ its membership in action t , with $m_{i,t,k} \leq q_{i,k}$ and $\sum_t m_{i,t,k} \leq 1$ so that local action spans do not overlap. Set $L_i = \sum_k q_{i,k}$ and $B_{i,t} = \sum_k m_{i,t,k}$. Actions with $B_{i,t} = 0$ contribute no local loss, while trajectories with $L_i = 0$ contribute no policy loss. For trajectories sampled from the same task, the group-relative outcome advantage is (Shao et al., 2024)

$$A_i^{\text{out}} = \frac{R_i - \mu_x}{\sigma_x + \varepsilon}, \quad (11)$$

where μ_x and σ_x are the mean and population standard deviation of the group rewards and $\varepsilon > 0$ is

Model		Overall score					
		BFCL-v3	BFCL-v4	τ -bench	τ^2 -Bench	ToolSandbox	ACEBench
4B	Qwen3 (Yang et al., 2025)	25.38	38.61	33.44	15.83	52.89	55.28
	EnvScaler (Song et al., 2026)	37.25	<u>52.92</u>	48.78	33.05	62.25	63.33
	Arctic-AWM (Wang et al., 2026b)	38.89	49.55	26.58	22.57	68.24	60.58
	ECHO (ours)	41.24	53.71	50.77	35.81	76.36	<u>62.46</u>
8B	Qwen3 (Yang et al., 2025)	28.88	41.58	36.65	26.44	59.74	60.00
	EnvScaler (Song et al., 2026)	41.38	54.20	44.22	35.10	<u>68.85</u>	58.33
	Arctic-AWM (Wang et al., 2026b)	<u>43.70</u>	52.50	37.64	33.45	60.44	54.93
	Klear-AgentForge (Wang et al., 2025a)	37.84	47.88	<u>49.10</u>	35.91	66.06	60.70
	MUA-RL (Zhao et al., 2025)	14.63	26.84	43.00	30.20	50.00	53.30
	Simia-RL (Li et al., 2025)	41.64	<u>55.72</u>	44.27	<u>40.16</u>	62.25	58.33
	ECHO (ours)	49.17	56.91	50.59	40.56	74.42	<u>60.08</u>

Table 1: Overall results on six tool-use benchmarks at the 4B and 8B scales, spanning multi-turn function calling, conversational task completion, and stateful tool execution. For BFCL-v3, we report the overall score on the Multi-Turn (MT) subset. For ACEBench, we report results on the Agent subset. Scores are percentages. **Bold** and underlined values mark the best and second-best results within each scale.

a small constant. Equal-reward groups therefore have zero outcome advantage but may still receive local corrections.

Let $\rho_{i,k}$ be the current-to-behavior token likelihood ratio, and define the clipped surrogate (Schulman et al., 2017)

$$\phi_\epsilon(\rho, A) = \min\{\rho A, \text{clip}(\rho, 1-\epsilon, 1+\epsilon)A\}. \quad (12)$$

ECHO maximizes

$$J_i^{\text{out}} = \frac{1}{L_i} \sum_k q_{i,k} \phi_\epsilon(\rho_{i,k}, A_i^{\text{out}}), \quad (13)$$

$$J_i^{\text{loc}} = \sum_{t: B_{i,t} > 0} \frac{1}{B_{i,t}} \sum_k m_{i,t,k} \phi_\epsilon(\rho_{i,k}, \delta_{i,t}), \quad (14)$$

$$J_{\text{ECHO}} = \frac{1}{N} \sum_i [J_i^{\text{out}} + J_i^{\text{loc}} - \beta K_i], \quad (15)$$

where K_i is the trajectory-mean reference-policy KL penalty and N counts contributing trajectories. Judge outputs, rewards, budgets, and advantages are fixed within each update.

The local branch averages over tokens within each action and then sums across actions, so $\delta_{i,t}$ is neither diluted by action length nor attenuated by clipping of length-rescaled token advantages. At the behavior policy, the split objective has the same gradient as the additive token advantage

$$A_i^{\text{out}} + L_i \sum_{t: B_{i,t} > 0} \frac{\delta_{i,t}}{B_{i,t}} m_{i,t,k}. \quad (16)$$

Once ratio clipping activates, opposite-sign outcome and local advantages may fall into different

clipping branches, so the split objective can differ from optimizing their combined advantage.

Source funding also bounds the local surrogate. For a fixed audited trajectory whose positive events are all self-funded, let $N_i^- = \sum_t \max\{-\delta_{i,t}, 0\}$. Then

$$J_i^{\text{loc}} \leq [(1+\epsilon)\eta - (1-\epsilon)]N_i^- \leq 0 \quad (17)$$

whenever $\eta \leq (1-\epsilon)/(1+\epsilon)$ and $0 < \epsilon < 1$. Under these conditions, repairing self-induced errors cannot yield a net positive local surrogate. This fixed-batch guarantee depends on correct attribution and does not by itself constrain learned behavior under shared policy parameters (§C.3). Optimization and cost details are given in §E.

4 Experiments

We evaluate ECHO along three dimensions. First, we assess its overall performance across six tool-use benchmarks at the 4B and 8B scales (§4.2). Second, we analyze the contributions of its design choices through component and reward ablations (§4.3 and §4.4). Finally, we examine how frozen-judge choice relates to responsibility identification and downstream policy performance (§4.5).

4.1 Setup

Benchmarks and metrics. We evaluate ECHO at the 4B and 8B scales on BFCL-v3 Multi-Turn, BFCL-v4, τ -bench, τ^2 -Bench, ToolSandbox, and ACEBench-Agent. BFCL evaluates function calling, with v4 additionally covering web search and memory use (Patil et al., 2025; Mao et al., 2025). We evaluate both τ -bench domains, retail and airline (Yao et al., 2024), and all three

Model		τ -bench			τ^2 -Bench			
		Retail	Airline	Overall	Retail	Airline	Telecom	Overall
4B	Qwen3 (Yang et al., 2025)	40.87	26.00	33.44	19.96	21.00	9.43	15.83
	EnvScaler (Song et al., 2026)	49.57	48.00	48.78	50.00	36.00	13.16	33.05
	Arctic-AWM (Wang et al., 2026b)	30.15	23.00	26.58	30.26	19.00	16.45	22.57
	ECHO (ours)	52.03	49.50	50.77	54.09	36.00	17.33	35.81
8B	Qwen3 (Yang et al., 2025)	45.79	27.50	36.65	34.43	26.50	18.42	26.44
	EnvScaler (Song et al., 2026)	50.43	38.00	44.22	<u>53.51</u>	36.00	15.79	35.10
	Arctic-AWM (Wang et al., 2026b)	39.88	35.40	37.64	41.23	38.50	<u>23.47</u>	33.45
	Klear-AgentForge (Wang et al., 2025a)	<u>56.70</u>	<u>41.50</u>	<u>49.10</u>	49.42	36.67	21.64	35.91
	MUA-RL (Zhao et al., 2025)	56.50	29.50	43.00	49.80	19.00	21.80	30.20
	Simia-RL (Li et al., 2025)	51.82	36.71	44.27	52.90	49.00	18.58	<u>40.16</u>
	ECHO (ours)	58.17	43.00	50.59	54.63	<u>42.10</u>	24.95	40.56

Table 2: Domain-level results on τ -bench and τ^2 -Bench at the 4B and 8B scales. All scores are reported as percentages. **Bold** and underlined values mark the best and second-best results within each model scale.

Model		Base	Missing Function	Missing Parameter	Long Context	Overall
4B	Qwen3 (Yang et al., 2025)	32.00	20.00	24.00	25.50	25.38
	EnvScaler (Song et al., 2026)	46.50	<u>38.50</u>	28.50	<u>35.50</u>	37.25
	Arctic-AWM (Wang et al., 2026b)	<u>48.55</u>	36.50	34.22	36.30	<u>38.89</u>
	ECHO (ours)	50.50	46.58	<u>32.59</u>	35.27	41.24
8B	Qwen3 (Yang et al., 2025)	32.00	33.50	22.00	28.00	28.88
	EnvScaler (Song et al., 2026)	50.50	41.50	29.50	<u>44.00</u>	41.38
	Arctic-AWM (Wang et al., 2026b)	53.40	<u>45.00</u>	32.40	<u>44.00</u>	<u>43.70</u>
	Klear-AgentForge (Wang et al., 2025a)	45.86	39.00	34.50	32.00	37.84
	MUA-RL (Zhao et al., 2025)	21.00	11.50	15.00	11.00	14.63
	Simia-RL (Li et al., 2025)	<u>54.50</u>	36.50	<u>37.54</u>	38.00	41.64
	ECHO (ours)	59.50	51.50	40.68	45.00	49.17

Table 3: BFCL-v3 Multi-Turn results across Base, Missing Function, Missing Parameter, and Long Context conditions at the 4B and 8B scales. Overall is the unweighted mean across the four conditions. All scores are reported as percentages. **Bold** and underlined values mark the best and second-best results within each model scale.

τ^2 -Bench domains, retail, airline, and telecom (Barres et al., 2025). These conversational benchmarks test policy-constrained task completion, with telecom additionally requiring coordinated agent-user actions. ToolSandbox evaluates intermediate and final state milestones (Lu et al., 2025), while ACEBench-Agent covers multi-step and multi-turn tool use in English and Chinese (Chen et al., 2025).

Baselines. Qwen3-4B and Qwen3-8B serve as the base policies at 4B and 8B, respectively (Yang et al., 2025). At 4B, we compare against EnvScaler (Song et al., 2026) and Arctic-AWM (Wang et al., 2026b). At 8B, we additionally compare against Klear-AgentForge (Wang et al., 2025a), MUA-RL (Zhao et al., 2025), and Simia-RL (Li et al., 2025).

Training and configuration. ECHO is trained from Qwen3-4B and Qwen3-8B on the τ^2 -Bench training split. Training details and hyperparameters are given in §D. The frozen-judge protocol and reliability gating are described in §A and §B.

4.2 Main Results

Overall performance. ECHO achieves the best result on five of six benchmarks at both model scales. At 4B, it outperforms the strongest same-scale baseline by 2.35 points on BFCL-v3 Multi-Turn, 0.79 on BFCL-v4, 1.99 on τ -bench, 2.76 on τ^2 -Bench, and 8.12 on ToolSandbox. At 8B, the corresponding gains are 5.47, 1.19, 1.49, 0.40, and 5.57 points. ACEBench-Agent is the only benchmark on which ECHO does not lead, trailing the best baseline by 0.87 points at 4B and 0.62 at 8B. The ranking pattern is consistent across scales: ECHO leads on the same five benchmarks at 4B and 8B, with the largest margins on BFCL-v3 Multi-Turn and ToolSandbox.

Domain-level results. ECHO leads or ties the strongest baseline on all five domain-level comparisons at 4B. At 8B, it leads τ -bench Retail and Airline by 1.47 and 1.50 points, and τ^2 -Bench Retail and Telecom by 1.12 and 1.48 points, respectively. The main exception is τ^2 -Bench Airline, where ECHO scores 42.10 versus 49.00 for Simia-RL.

Despite this 6.90-point deficit, ECHO achieves the highest τ^2 -Bench overall score, 40.56 versus 40.16 for Simia-RL, as its gains on Retail and Telecom outweigh the Airline deficit.

BFCL-v3 condition-level results. At 4B, ECHO leads on Base and Missing Function by 1.95 and 8.08 points, respectively, but trails the strongest baseline by 1.63 points on Missing Parameter and 1.03 on Long Context. At 8B, ECHO leads all four conditions, with gains of 5.00 points on Base, 6.50 on Missing Function, 3.14 on Missing Parameter, and 1.00 on Long Context. This advantage extends to all four conditions at 8B, whereas at 4B it is concentrated in Base and Missing Function.

4.3 Component Ablation

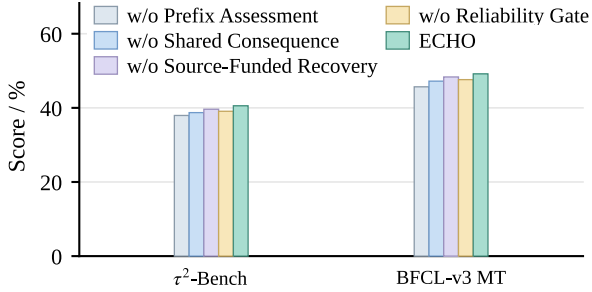


Figure 3: Ablation of ECHO components on τ^2 -Bench and BFCL-v3 Multi-Turn at the 8B scale.

Figure 3 presents one-component-at-a-time ablations of ECHO. w/o Prefix Assessment omits independent prefix assessment; w/o Shared Consequence assigns full consequence mass to each event; w/o Source-Funded Recovery removes source-funded recovery constraints; and w/o Reliability Gate disables calibrated semantic filtering. All four ablations reduce performance on both benchmarks, with a consistent ordering of drops. Removing prefix assessment causes the largest degradation, suggesting a central role for decision-time responsibility, whereas removing source-funded recovery has the smallest effect.

4.4 Reward Ablation

At the 8B scale, we compare ECHO with Final Reward Only, Random One Error, First Error, and All Errors. Final Reward Only retains terminal supervision, while the three heuristics penalize one random error, the first error, or all identified errors, respectively. ECHO exceeds Final Reward Only by 9.51 points and First Error, the strongest heuristic, by 2.77 points overall. Although All Errors and First Error lead Airline and Telecom, respectively,

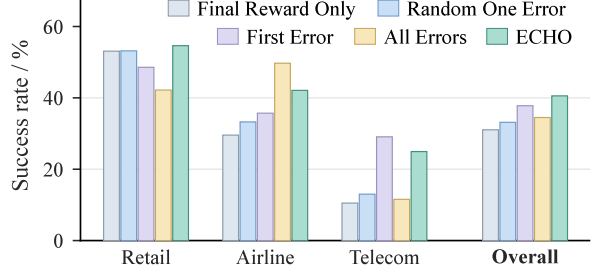


Figure 4: Reward ablation on τ^2 -Bench at the 8B scale. Each group compares five supervision configurations, and Overall denotes the mean across the three domains.

ECHO achieves the highest Retail score and the highest overall score. These results suggest that ECHO benefits not simply from additional local supervision, but from responsibility-aware credit assignment across errors and corrections. §F reports the same comparison on BFCL-v3 Multi-Turn.

4.5 Effect of the Frozen Judge

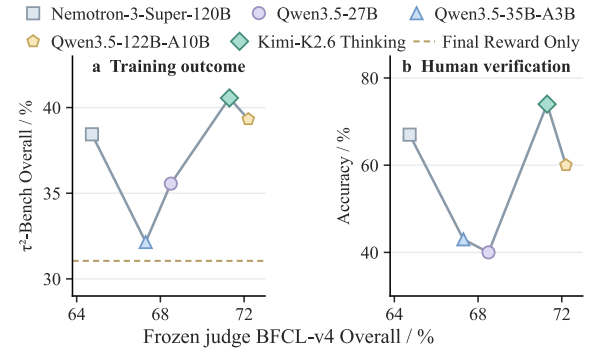


Figure 5: Frozen-judge comparison at the 8B scale. Left: downstream ECHO performance on τ^2 -Bench. Right: responsibility identification accuracy on 100 manually verified samples before reliability gating.

Frozen-judge comparisons examine whether general tool-use capability predicts effectiveness as a source of local supervision. Figure 5 compares five judges by BFCL-v4 score, downstream ECHO-8B performance, and responsibility identification accuracy on 100 manually verified samples. Nemotron-3-Super-120B has the lowest BFCL-v4 score, yet yields stronger downstream ECHO performance than both smaller Qwen judges. Kimi-K2.6 Thinking produces the strongest ECHO policy, exceeding Qwen3.5-122B-A10B by 1.24 percentage points despite the latter’s higher BFCL-v4 score. These reversals suggest that general tool-use capability is an incomplete proxy for supervision effectiveness.

5 Conclusion

We presented ECHO, a responsibility-aware credit assignment framework for tool-use agents that dis-

tinguishes newly introduced errors, inherited consequences, missed corrections, and successful corrections, and converts them into bounded action-level supervision. This design addresses three limitations of conventional terminal or step-level feedback: (i) conflating error manifestations with responsible decisions, (ii) double-counting shared downstream consequences, and (iii) over-crediting recovery from self-induced errors. Across six tool-use benchmarks and two model scales, ECHO achieves strong and broadly consistent improvements over the compared baselines. Together, these results support decision responsibility as a useful perspective for fine-grained credit assignment across complex tool-use trajectories.

Limitations

This work focuses specifically on responsibility-aware credit assignment for tool-use agents. (i) ECHO is designed for training-time credit assignment rather than inference-time intervention. Techniques such as reflection, replanning, and online verification are complementary to the proposed framework and are not considered here. (ii) Our analysis centers on responsibility attribution within agent–tool interaction trajectories. Broader system-level considerations, including tool construction, multi-agent coordination, and interface design, fall outside the scope of the proposed framework.

Ethical Considerations

All experiments in this study were conducted using publicly available datasets and models licensed for research use. To the best of our knowledge, we do not identify any method-specific ethical concerns beyond those generally associated with tool-use agents and automated model evaluation.

References

- Jose A Arjona-Medina, Michael Gillhofer, Michael Widrich, Thomas Unterthiner, Johannes Brandstetter, and Sepp Hochreiter. 2019. Rudder: Return decomposition for delayed rewards. In *Advances in Neural Information Processing Systems*. Available: <https://arxiv.org/pdf/1806.07857>.
- Victor Barres, Honghua Dong, Soham Ray, Xujie Si, and Karthik Narasimhan. 2025. τ^2 -bench: Evaluating conversational agents in a dual-control environment. *arXiv preprint arXiv:2506.07982*. Available: <https://arxiv.org/pdf/2506.07982>.
- Chen Chen, Xinlong Hao, Weiwen Liu, Xu Huang, Xingshan Zeng, Shuai Yu, Dexun Li, Shuai Wang, Weinan Gan, Yuefeng Huang, et al. 2025. Acebench: Who wins the match point in tool usage? *arXiv preprint arXiv:2501.12851*. Available: <https://arxiv.org/pdf/2501.12851>.
- Hana Chockler and Joseph Y Halpern. 2004. Responsibility and blame: A structural-model approach. *Journal of Artificial Intelligence Research*. Available: <https://arxiv.org/pdf/cs/0312038>.
- Guanting Dong, Hangyu Mao, Kai Ma, Licheng Bao, Yifei Chen, Zhongyuan Wang, Zhongxia Chen, Jiazhen Du, Huiyang Wang, Fuzheng Zhang, et al. 2026. Agentic reinforced policy optimization. In *International Conference on Learning Representations*. Available: <https://arxiv.org/pdf/2507.19849>.
- Jiazhan Feng, Shijue Huang, Xingwei Qu, Ge Zhang, Yujia Qin, Baoquan Zhong, Chengquan Jiang, Jinxin Chi, and Wanjun Zhong. 2026a. Retool: Reinforcement learning for strategic tool use in llms. In *International Conference on Learning Representations*. Available: <https://arxiv.org/pdf/2504.11536>.
- Lang Feng, Zhenghai Xue, Tingcong Liu, and Bo An. 2026b. Group-in-group policy optimization for llm agent training. In *Advances in Neural Information Processing Systems*. Available: <https://arxiv.org/pdf/2505.10978>.
- Shubham Gandhi, Saurabh Goyal, Kiran Kate, and Yara Rizk. 2026. Draco: Fine-grained credit assignment with dynamic rubrics for long-horizon agent training. *arXiv preprint arXiv:2609.04094*. Available: <https://arxiv.org/pdf/2609.04094>.
- Jiazheng Li, Yawei Wang, Qiaojing Yan, Yijun Tian, Zhichao Xu, Huan Song, Panpan Xu, and Lin Lee Cheong. 2026a. Salt: Step-level advantage assignment for long-horizon agents via trajectory graph. In *Findings of the Association for Computational Linguistics: EACL 2026*. Available: <https://aclanthology.org/2026.findings-eacl.247.pdf>.
- Renhao Li, Jianhong Tu, Yang Su, Yantao Liu, Fei Huang, Hamid Alinejad-Rokny, Derek F Wong, Junyang Lin, and Min Yang. 2026b. Toolrm: Towards agentic tool-use reward modeling. In *Findings of the Association for Computational Linguistics: ACL 2026*. Available: <https://aclanthology.org/2026.findings-acl.419.pdf>.
- Yuetai Li, Huseyin A Inan, Xiang Yue, Wei-Ning Chen, Lukas Wutschitz, Janardhan Kulkarni, Radha Pooven-dran, Robert Sim, and Saravan Rajmohan. 2025. Simulating environments with reasoning models for agent training. *arXiv preprint arXiv:2511.01824*. Available: <https://arxiv.org/pdf/2511.01824>.
- Sicong Liao, Zhi Chen, and Yaohua Tang. 2026. Tcpo: Turn-level credit policy optimization. *arXiv preprint arXiv:2608.01667*. Available: <https://arxiv.org/pdf/2608.01667>.
- Hunter Lightman, Vineet Kosaraju, Yuri Burda, Harrison Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. 2024. Let’s verify step by step. In *International Conference on Learning Representations*. Available: <https://arxiv.org/pdf/2305.20050>.
- Jiarui Lu, Thomas Holleis, Yizhe Zhang, Bernhard Aumayer, Feng Nan, Haoping Bai, Shuang Ma, Shen Ma, Mengyu Li, Guoli Yin, et al. 2025. Toolsand-box: A stateful, conversational, interactive evaluation benchmark for llm tool use capabilities. In *Findings of the Association for Computational Linguistics: NAACL 2025*. Available: <https://aclanthology.org/2025.findings-naacl.65.pdf>.
- Xufang Luo, Yuge Zhang, Zhiyuan He, Zilong Wang, Siyun Zhao, Dongsheng Li, Luna K Qiu, and Yuqing Yang. 2025. Agent lightning: Train any ai agents with reinforcement learning. *arXiv preprint*

- arXiv:2508.03680. Available: <https://arxiv.org/pdf/2508.03680>.
- Huanzhi Mao, Raymond Tsao, Jingzhuo Zhou, Shishir G. Patil, and Joseph E. Gonzalez. 2025. *BFCL V4: Web search*. Berkeley Gorilla project.
- Zibin Meng and Kani Chen. 2026. Craft: Counterfactual credit assignment from free sibling rollouts for self-distilled agentic reinforcement learning. *arXiv preprint arXiv:2606.29476*. Available: <https://arxiv.org/pdf/2606.29476>.
- Alexander Meulemans, Simon Schug, Seijin Kobayashi, Gregory Wayne, et al. 2023. Would i have gotten that reward? long-term credit assignment by counterfactual contribution analysis. In *Advances in Neural Information Processing Systems*. Available: <https://arxiv.org/pdf/2306.16803>.
- Shishir G Patil, Huanzhi Mao, Fanjia Yan, Charlie Cheng-Jie Ji, Vishnu Suresh, Ion Stoica, and Joseph E Gonzalez. 2025. The berkeley function calling leaderboard (bfcl): From tool use to agentic evaluation of large language models. In *International Conference on Machine Learning*. Available: <https://proceedings.mlr.press/v267/patil25a.html>.
- Cheng Qian, Emre Can Acikgoz, Qi He, Hongru Wang, Xiushi Chen, Dilek Hakkani-Tur, Gokhan Tur, and Heng Ji. 2026. Toolrl: Reward is all tool learning needs. In *Advances in Neural Information Processing Systems*. Available: <https://arxiv.org/pdf/2504.13958>.
- Yujia Qin, Shihao Liang, Yining Ye, Kunlun Zhu, Lan Yan, Yaxi Lu, Yankai Lin, Xin Cong, Xiangru Tang, Bill Qian, et al. 2024. Toolllm: Facilitating large language models to master 16000+ real-world apis. In *International Conference on Learning Representations*. Available: <https://arxiv.org/pdf/2307.16789>.
- Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Eric Hambro, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. 2023. Toolformer: Language models can teach themselves to use tools. In *Advances in Neural Information Processing Systems*. Available: <https://arxiv.org/pdf/2302.04761>.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*. Available: <https://arxiv.org/pdf/1707.06347>.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, YK Li, Yang Wu, et al. 2024. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*. Available: <https://arxiv.org/pdf/2402.03300>.
- Xiaoshuai Song, Haofei Chang, Guanting Dong, Yutao Zhu, Ji-Rong Wen, and Zhicheng Dou. 2026. Envscler: Scaling tool-interactive environments for llm agent via programmatic synthesis. In *Findings of the Association for Computational Linguistics: ACL 2026*. Available: <https://aclanthology.org/2026.findings-acl.407.pdf>.
- Hui-Ze Tan, Xiao-Wen Yang, Hao Chen, Jie-Jing Shao, Yi Wen, Yuteng Shen, Weihong Luo, Xiku Du, Lan-Zhe Guo, and Yu-Feng Li. 2026. Hind-sight credit assignment for long-horizon llm agents. *arXiv preprint arXiv:2603.08754*. Available: <https://arxiv.org/pdf/2603.08754>.
- Weiting Tan, Xinghua Qu, Ming Tu, Meng Ge, Andy T Liu, Philipp Koehn, and Lu Lu. 2025. Process-supervised reinforcement learning for interactive multimodal tool-use agents. *arXiv preprint arXiv:2509.14480*. Available: <https://arxiv.org/pdf/2509.14480>.
- Leitian Tao, Baolin Peng, Wenlin Yao, Tao Ge, Hao Cheng, Mike Hang Wang, Jianfeng Gao, and Sharon Li. 2026. Trace: Turn-level reward assignment via credit estimation for long-horizon agents. *arXiv preprint arXiv:2607.13988*. Available: <https://arxiv.org/pdf/2607.13988>.
- Junzhe Wang, Zhiheng Xi, Yajie Yang, Hao Luo, Shihan Dou, Tao Gui, and Qi Zhang. 2026a. Enhancing llm-based search agents via contribution weighted group relative policy optimization. In *Annual Meeting of the Association for Computational Linguistics*. Available: <https://aclanthology.org/2026.acl-long.1462.pdf>.
- Peiyi Wang, Lei Li, Zhihong Shao, Runxin Xu, Damai Dai, Yifei Li, Deli Chen, Yu Wu, and Zhifang Sui. 2024. Math-shepherd: Verify and reinforce llms step-by-step without human annotations. In *Annual Meeting of the Association for Computational Linguistics*. Available: <https://aclanthology.org/2024.acl-long.510.pdf>.
- Qi Wang, Hongzhi Zhang, Jia Fu, Kai Fu, Yahui Liu, Tinghai Zhang, Chenxi Sun, Gangwei Jiang, Jingyi Tang, Xingguang Ji, et al. 2025a. Klear-agentforge: Forging agentic intelligence through posttraining scaling. *arXiv preprint arXiv:2511.05951*. Available: <https://arxiv.org/pdf/2511.05951>.
- Zhaoyang Wang, Canwen Xu, Boyi Liu, Yite Wang, Siwei Han, Zhewei Yao, Huaxiu Yao, and Yuxiong He. 2026b. Agent world model: Infinity synthetic environments for agentic reinforcement learning. *arXiv preprint arXiv:2602.10090*. Available: <https://arxiv.org/pdf/2602.10090>.
- Zihan Wang, Kangrui Wang, Qineng Wang, Pingyue Zhang, Linjie Li, Zhengyuan Yang, Xing Jin, Kefan Yu, Minh Nhat Nguyen, Licheng Liu, et al. 2025b. Ragen: Understanding self-evolution in llm agents via multi-turn reinforcement learning. *arXiv preprint arXiv:2504.20073*. Available: <https://arxiv.org/pdf/2504.20073>.

- Muning Wen, Ziyu Wan, Jun Wang, Weinan Zhang, and Ying Wen. 2024. Reinforcing llm agents via policy optimization with action decomposition. In *Advances in Neural Information Processing Systems*. Available: <https://arxiv.org/pdf/2405.15821>.
- Zhiheng Xi, Chenyang Liao, Guanyu Li, Zhihao Zhang, Wenxiang Chen, Binghai Wang, Senjie Jin, Yuhao Zhou, Jian Guan, Wei Wu, et al. 2026. Agentprm: Process reward models for llm agents via step-wise promise and progress. In *The ACM Web Conference 2026*. Available: <https://arxiv.org/pdf/2511.08325>.
- Yu Xia, Jingru Fan, Weize Chen, Siyu Yan, Xin Cong, Zhong Zhang, Yaxi Lu, Yankai Lin, Zhiyuan Liu, and Maosong Sun. 2025. Agentrm: Enhancing agent generalization with reward modeling. In *Annual Meeting of the Association for Computational Linguistics*. Available: <https://aclanthology.org/2025.acl-long.945.pdf>.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*. Available: <https://arxiv.org/pdf/2505.09388>.
- Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. 2024. τ -bench: A benchmark for tool-agent-user interaction in real-world domains. *arXiv preprint arXiv:2406.12045*. Available: <https://arxiv.org/pdf/2406.12045>.
- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2022. React: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629*. Available: <https://arxiv.org/pdf/2210.03629>.
- Yan Zhan, Shaobo Liu, Qiunan Liu, Yuanjun Shi, Siqi Xu, WeiYi Hou, Xiang Xu, Zekang Li, Weizhou Pan, and Jiahong Yan. 2026. Slca-grpo: Resolving cross-segment credit misattribution in tool-calling rl. *arXiv preprint arXiv:2609.29050*. Available: <https://arxiv.org/pdf/2609.29050>.
- Shaokun Zhang, Ming Yin, Jieyu Zhang, Jiale Liu, Zhiguang Han, Jingyang Zhang, Beibin Li, Chi Wang, Huazheng Wang, Yiran Chen, et al. 2025a. Which agent causes task failures and when? on automated failure attribution of llm multi-agent systems. *arXiv preprint arXiv:2505.00212*. Available: <https://arxiv.org/pdf/2505.00212>.
- Yaocheng Zhang, Haohuan Huang, Zijun Song, Zijie Zhao, Qichao Zhang, Yuanheng Zhu, and Dongbin Zhao. 2026. Criticsearch: Fine-grained credit assignment for search agents via a retrospective critic. In *Findings of the Association for Computational Linguistics: ACL 2026*. Available: <https://aclanthology.org/2026.findings-acl.596.pdf>.
- Yudi Zhang, Yali Du, Biwei Huang, Ziyang Wang, Jun Wang, Meng Fang, and Mykola Pechenizkiy. 2023. Interpretable reward redistribution in reinforcement learning: A causal approach. In *Advances in Neural Information Processing Systems*. Available: <https://arxiv.org/pdf/2305.18427>.
- Zhenru Zhang, Chujie Zheng, Yangzhen Wu, Beichen Zhang, Runji Lin, Bowen Yu, Dayiheng Liu, Jingren Zhou, and Junyang Lin. 2025b. The lessons of developing process reward models in mathematical reasoning. In *Findings of the Association for Computational Linguistics: ACL 2025*. Available: <https://aclanthology.org/2025.findings-acl.547.pdf>.
- Weikang Zhao, Xili Wang, Chengdi Ma, Lingbin Kong, Zhaohua Yang, Mingxiang Tuo, Xiaowei Shi, Yitao Zhai, and Xunliang Cai. 2025. Mua-rl: Multi-turn user-interacting agent reinforcement learning for agentic tool use. *arXiv preprint arXiv:2508.18669*. Available: <https://arxiv.org/pdf/2508.18669>.
- Chujie Zheng, Zhenru Zhang, Beichen Zhang, Runji Lin, Keming Lu, Bowen Yu, Dayiheng Liu, Jingren Zhou, and Junyang Lin. 2025. Processbench: Identifying process errors in mathematical reasoning. In *Annual Meeting of the Association for Computational Linguistics*. Available: <https://aclanthology.org/2025.acl-long.50.pdf>.
- Shuyan Zhou, Frank F Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng, Tianyue Ou, Yonatan Bisk, Daniel Fried, et al. 2024a. Webarena: A realistic web environment for building autonomous agents. In *International Conference on Learning Representations*. Available: <https://arxiv.org/pdf/2307.13854>.
- Yifei Zhou, Song Jiang, Yuandong Tian, Jason Weston, Sergey Levine, Sainbayar Sukhbaatar, and Xian Li. 2025. Sweet-rl: Training multi-turn llm agents on collaborative reasoning tasks. *arXiv preprint arXiv:2503.15478*. Available: <https://arxiv.org/pdf/2503.15478>.
- Yifei Zhou, Andrea Zanette, Jiayi Pan, Sergey Levine, and Aviral Kumar. 2024b. Archer: Training language model agents via hierarchical multi-turn rl. *arXiv preprint arXiv:2402.19446*. Available: <https://arxiv.org/pdf/2402.19446>.
- Kunlun Zhu, Zijia Liu, Bingxuan Li, Muxin Tian, Yingxuan Yang, Jiaxun Zhang, Pengrui Han, Qipeng Xie, Fuyang Cui, Weijia Zhang, et al. 2025. Where llm agents fail and how they can learn from failures. *arXiv preprint arXiv:2509.25370*. Available: <https://arxiv.org/pdf/2509.25370>.

Contents of Appendix

A Audit Protocol and Evidence Handling	13
A.1 Candidate Selection and Independent Requests	13
A.2 Responsibility Boundaries	13
B Calibration of Selective Supervision	14
C Derivations and Guarantee Boundaries	14
C.1 Shared-Consequence and Budget Bounds	14
C.2 First-Order Equivalence	14
C.3 Source Funding and the Clipped Local Surrogate	15
C.4 Observational Attribution Limits	15
D Implementation Details	15
D.1 Training Data and Implementation	15
D.2 ECHO Configuration	15
E Optimization Details and Computational Cost	16
E.1 Update Sequence	16
E.2 Audit Cost	16
E.3 Realized Local Coefficients	16
F Reward Ablation on Multi-Turn Function Calling	16

A Audit Protocol and Evidence Handling

A.1 Candidate Selection and Independent Requests

The full-trajectory proposer returns at most sixteen event hypotheses and explicitly marks omitted records. Each hypothesis identifies its action, policy span, versioned predicate, deviation origin, evidence pointers, and proposed responsibility type. Impact records specify propagation links, severity, first-restoration evidence, and authenticated origins. Validated fields determine training weights through the scoring rules in §3.3–3.4.

The prefix-review budget covers $K = 8$ distinct actions. Up to six proposed actions are selected chronologically; two non-proposed sentinel actions follow a fixed-seed hash ordering. This ordering also fills unused proposal slots until the budget or available actions are exhausted. Each request returns at most four predicate-level records, with omitted records assigned abstention.

Each prefix request contains the recorded decision-time context $h_{i,t}$, current action, and avail-

able tool contracts. Retrospective labels, selected predicates, rationales, severity, suffixes, and hidden state are excluded from these inputs. Batched requests retain isolated attention contexts, and both inference roles use the same frozen checkpoint with fixed prompts and decoding. Prefix assessment returns supported defects, legitimate repair intentions, defect-free assessments, or abstention. The assembler combines an accepted repair intention with execution confirmation to establish a successful correction.

Sentinel-discovered defects contribute through the direct term when their prefix evidence passes validation. Unsupported impact and propagation terms receive zero weight, and sentinel repairs require authenticated restoration evidence for positive supervision.

A.2 Responsibility Boundaries

Inherited inputs and current obligations. The classification of a reused product identifier depends on the discrepancy evidence visible when it is used. A visible mismatch and an available resolving lookup can establish a missed correction when the agent continues committing to the wrong product. An apparently valid opaque identifier can carry an inherited consequence when its defect becomes observable later. Ambiguous cases receive local abstention, preserving the distinction between available evidence and retrospectively established error.

Feasible responses. The feasible alternative response (witness) of a missed correction specifies a concrete tool or response, known arguments, availability under the visible contract, and the discrepancy addressed. Clarification qualifies when the user can disambiguate exposed alternatives; stopping qualifies when permitted and sufficient to avoid the identified impermissible operation. A legitimate repair attempt retains its prefix-grounded assessment even if an unpredictable outage later prevents restoration.

Mixed actions and verified restoration. An action that restores one field while violating another produces distinct positive and negative events. Transaction identifiers or field-level provenance link verified restoration to the responsible action. Ambiguous restoration attribution triggers abstention from positive supervision. A verified correction remains valid despite a separate terminal failure, with auxiliary eligibility determined by its origin and available source funding.

Visibility and external origins. Decision-time visibility follows delivery timestamps and actual context inclusion, placing synchronous tool output after its generating action. Hidden snapshots support retrospective impairment assessment, while correction opportunities require discrepancy evidence visible in $h_{i,t}$. Initial-state records authenticate pre-existing defects, and environment records authenticate subsequent external incidents. Omitted sources and unsupported policy claims leave origins unknown. A new policy error that reopens a repaired predicate creates a separate self-induced origin with its own source penalty.

B Calibration of Selective Supervision

Prompt development, gate calibration, hyperparameter selection, and final evaluation use disjoint task partitions. A fixed selection rule samples one eligible event per independent task instance within each predeclared evidence stratum. Events and rollouts sharing a task form one sampling cluster. Two annotators assess prefix-grounded responsibility, followed by a separate retrospective review of impact, provenance, and restoration. Adjudication is blinded to training coefficients, and correctness requires validity of every training-relevant semantic field.

For stratum b , let k_b of n_b independent calibration units contain an incorrect assertion. With B_{cal} predeclared strata and simultaneous failure level δ_{cal} , define

$$U_b = \text{Beta}^{-1} \left[1 - \frac{\delta_{\text{cal}}}{B_{\text{cal}}}; k_b + 1, n_b - k_b \right] \quad (18)$$

for $0 \leq k_b < n_b$, and set $U_b = 1$ if $k_b = n_b$ or $n_b = 0$. The inverse beta distribution yields a one-sided exact binomial upper confidence limit. Combining it with the indicator V_e of the structural checks in §3.2 gives

$$g_e = V_e \mathbf{1}\{n_{b(e)} \geq n_{\min}\} \mathbf{1}\{U_{b(e)} \leq \epsilon_J\}. \quad (19)$$

Independent sampling from the same stratum distributions gives simultaneous upper coverage of at least $1 - \delta_{\text{cal}}$ through a union bound. This guarantee concerns stratum-level semantic-error rates under the calibration distribution. Individual-event validity, impact magnitude, and reliability under policy shift remain separate assessment targets. Post hoc stratum selection uses an independent selection split to preserve the calibration protocol.

The calibration design crosses three supervised responsibility types with explicit or semantic evidence to form six strata. Thresholds are $\delta_{\text{cal}} = 0.05$, $\epsilon_J = 0.10$, and $n_{\min} = 50$; insufficiently sampled or unreliable strata abstain. Precision and coverage jointly characterize selective supervision, while monitoring newly annotated training traces supports drift detection and recalibration on fresh development data.

C Derivations and Guarantee Boundaries

C.1 Shared-Consequence and Budget Bounds

Equation (3) gives $\sum_e a_{ec} = 1 - \zeta_c \leq 1$ for nonempty P_c , and zero otherwise. Let v_e^{shared} denote the shared-consequence component of a negative event's weight. Since $g_e \leq 1$ and $p_e \leq 1$,

$$\begin{aligned} \sum_e v_e^{\text{shared}} &= (1 - \kappa) \sum_c M_c \sum_e a_{ec} g_e [1 + \alpha p_e] \\ &\leq (1 - \kappa)(1 + \alpha) \sum_c M_c. \end{aligned} \quad (20)$$

The shared-consequence bound is independent of the number of implicated events; direct-defect terms separately supervise certified decisions. Duplicating observations preserves the accounting when validated predicates, event keys, provenance links, and first-restoration sets are fixed. The invariance therefore concerns validated records, with proposer sensitivity to redundant text requiring empirical assessment.

Equation (7) caps each directional total at λ_s . Source funding in Eq. (9) only reduces or preserves positive budgets, giving

$$\sum_t |\delta_{i,t}| \leq \sum_e b_e^- + \sum_r \tilde{b}_r^+ \leq \lambda_- + \lambda_+. \quad (21)$$

The resulting bound limits total auxiliary budget, while additional responsibility events can reduce individual normalized shares.

C.2 First-Order Equivalence

At the behavior policy, likelihood ratios equal one and lie inside the clipping interval. Writing the token score gradient as $s_{i,k} = \nabla_{\theta} \log \pi_{\theta}[y_{i,k} | h_{i,k}]$, where $h_{i,k}$ is the token-level context, the uncapped

additive-token surrogate gives

$$\begin{aligned}\nabla J_i^{\text{add}} &= \frac{A_i^{\text{out}}}{L_i} \sum_k q_{i,k} s_{i,k} \\ &\quad + \sum_{t: B_{i,t} > 0} \frac{\delta_{i,t}}{B_{i,t}} \sum_k m_{i,t,k} s_{i,k} \\ &= \nabla[J_i^{\text{out}} + J_i^{\text{loc}}].\end{aligned}\quad (22)$$

Objective values also agree whenever all relevant ratios lie strictly inside the clipping interval. Outside it, opposite-sign advantages can activate different clipping branches, yielding distinct objectives with an identical KL term.

If each token score gradient has norm at most G_0 at the behavior policy, then

$$\|\nabla J_i^{\text{loc}}\| \leq G_0 \sum_t |\delta_{i,t}| \leq G_0[\lambda_- + \lambda_+]. \quad (23)$$

This gradient bound holds under the score-gradient assumption at the sampling policy. Extending it across tokenizations or away from that policy requires additional assumptions.

C.3 Source Funding and the Clipped Local Surrogate

For a fixed audited trajectory with exclusively self-funded positive events, define gross budgets $P_0 = \sum_r \tilde{b}_r^+$ and $N_0 = \sum_e b_e^-$. Each source penalty D_d in Eq. (8) sums a disjoint subset of negative charges, and paid repairs of origin d total at most ηD_d , yielding

$$P_0 \leq \eta \sum_d D_d \leq \eta N_0. \quad (24)$$

Let C be the amount canceled when positive and negative events share actions. Net masses $P = P_0 - C$ and $N = N_0 - C$, where $N = N_i^-$ as defined in Eq. (17), satisfy

$$P - \eta N = P_0 - \eta N_0 - (1 - \eta)C \leq 0 \quad (25)$$

for $0 \leq \eta \leq 1$. For any $\rho > 0$ and $a, b \geq 0$, the clipped surrogate obeys

$$\phi_\epsilon(\rho, a) \leq (1 + \epsilon)a, \quad (26)$$

$$\phi_\epsilon(\rho, -b) \leq -(1 - \epsilon)b. \quad (27)$$

Averaging over each action’s tokens preserves these inequalities, yielding

$$\begin{aligned}J_i^{\text{loc}} &\leq (1 + \epsilon)P - (1 - \epsilon)N \\ &\leq [(1 + \epsilon)\eta - (1 - \epsilon)]N.\end{aligned}\quad (28)$$

This establishes Eq. (17) under its stated condition. Within trajectories containing external bonuses, the bound applies to self-induced components with action-disjoint support or independently evaluated local surrogates. These conditions preserve component-level accounting under nonlinear clipping.

Shared policy parameters can increase error creation despite the scalar source-funding bound. Consider an erroneous action with probability $p[\theta] = \sigma[\theta]$ and a subsequent repair with probability $q[\theta] = \sigma[H\theta]$. Assign coefficients $-D$ to the source and ηD to its verified repair. At $\theta = 0$, the expected local score-gradient update is

$$g_{\text{loc}}[0] = -\frac{D}{4} + \frac{\eta DH}{8}. \quad (29)$$

For $H > 2/\eta$, the update increases error probability while satisfying the source-funding constraint. The counterexample separates a bounded local surrogate from behavioral guarantees and motivates empirical assessment of error introduction, correction, and repeated repair cycles.

C.4 Observational Attribution Limits

A failed trajectory can support causal models with different intervention effects. For binary errors E_1, E_2 , both $F = E_1 \vee E_2$ and $F = E_1 \wedge E_2$ explain $E_1 = E_2 = F = 1$. Correcting E_1 preserves failure under the first model and resolves it under the second. The trace therefore leaves intervention-based responsibility underdetermined. ECHO uses the ownership convention in Eq. (3) to allocate observed constraint impairment.

D Implementation Details

D.1 Training Data and Implementation

ECHO is initialized from Qwen3-4B and Qwen3-8B and trained through reinforcement learning. We use 92 tasks from the τ^2 -Bench training splits, comprising 66 retail and 26 airline tasks. A further 8 retail and 4 airline tasks form the 12-task development split. Both scales use 75 updates, 12 tasks per batch, eight rollouts per task, and a learning rate of 5×10^{-6} .

D.2 ECHO Configuration

Kimi-K2.6 Thinking serves as the fixed judge for retrospective proposals and independent prefix assessments. Each trajectory receives one proposal request and at most eight prefix assessments, with

six proposal slots and two sentinel slots. We use $\kappa = 0.25$, $\alpha = 0.5$, $\lambda_- = \lambda_+ = 0.5$, $\eta = 0.5$, and clipping half-width $\epsilon = 0.2$. Outcome advantages use population-standard-deviation normalization with $\varepsilon = 10^{-6}$. §A and §B specify evidence validation and reliability gating.

E Optimization Details and Computational Cost

E.1 Update Sequence

Algorithm 1 summarizes one update. Each update audits successful and failed trajectories and retains every group reward for outcome normalization. Negative budgets determine source penalties before eligible corrections are normalized and capped by source funding. Aggregated action coefficients remain fixed during optimization of Eq. (15); the judge operates during training. The coefficients in §D satisfy the source-funding condition $\eta \leq [1 - \epsilon]/[1 + \epsilon]$.

Stored behavior probabilities correspond to the rollout distribution, including its temperature and token support. The sampling convention uses temperature one with full token support, and distributed accumulation averages over contributing trajectories across workers.

E.2 Audit Cost

Auditing a trajectory uses one proposal request and at most K independent prefix requests. For input lengths n and output lengths m , the token cost is

$$C_i^{\text{judge}} = n_i^{\text{full}} + m_i^{\text{full}} + \sum_{t \in S_i} [n_{i,t}^{\text{pre}} + m_{i,t}^{\text{pre}}] + C_i^{\text{retry}}. \quad (30)$$

Requirement-inventory compilation is amortized per task and accounted for separately. Uncached prefix input scales up to $O(Kn_i)$; exact-prefix caching can reduce repeated computation, while batching preserves request count. Inference cost also depends on context length and attention implementation.

E.3 Realized Local Coefficients

Away from clipping boundaries, an active local token contributes $\delta_{i,t}\rho_{i,k}/B_{i,t}$ to its score-gradient

coefficient. Define

$$\chi_{i,t,k} = \mathbf{1}\{\delta_{i,t} > 0, \rho_{i,k} \leq 1 + \epsilon\} + \mathbf{1}\{\delta_{i,t} < 0, \rho_{i,k} \geq 1 - \epsilon\}, \quad (31)$$

$$\delta_{i,t}^{\text{eff}} = \frac{\delta_{i,t}}{B_{i,t}} \sum_k m_{i,t,k} \rho_{i,k} \chi_{i,t,k}. \quad (32)$$

Active negative-advantage ratios can exceed the upper clipping threshold, so realized coefficient magnitudes can exceed their on-policy masses. Realized coefficients and sign-specific clipping fractions characterize how supervision changes through aggregation and clipping. Gradient norms and local-to-outcome alignment further distinguish update magnitude from directional agreement.

F Reward Ablation on Multi-Turn Function Calling

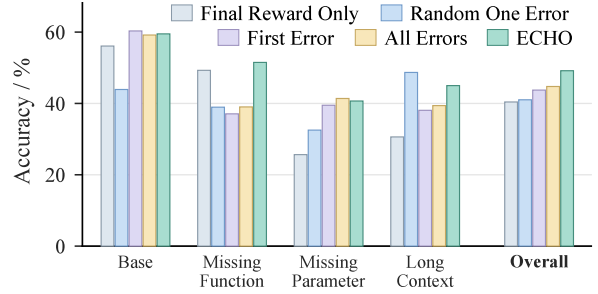


Figure 6: Reward ablation on BFCL-v3 Multi-Turn at 8B. Overall is the mean across Base, Missing Function, Missing Parameter, and Long Context.

The BFCL-v3 Multi-Turn ablation tests supervision under distinct information constraints, as shown in Figure 6. ECHO exceeded Final Reward Only by 8.76 percentage points and All Errors, the strongest ablation, by 4.42 points overall. Its advantage over All Errors was concentrated in Missing Function, where all three error-selection heuristics fell below terminal supervision. ECHO improved this subset while retaining gains over Final Reward Only in Missing Parameter and Long Context.

Subset leaders varied: First Error led Base, All Errors led Missing Parameter, and Random One Error led Long Context. The Long Context advantage of Random One Error accompanied a substantial Base deficit, illustrating the uneven effects of unstructured selection. The strongest heuristic also changed from First Error on τ^2 -Bench to All Errors on BFCL-v3 Multi-Turn. ECHO’s aggregate lead across both benchmarks supports its balance across decision conditions, alongside the observed advantages of simpler rules on individual subsets.